DIRAC EXPONENTIAL PARAMETRIZATION OF CONSTANT-RANK SKEW-SYMMETRIC FORMS

FLORIAN SCHAETZ, MARCO ZAMBON

Allowed standard prerequisites (editorial).

- Research prerequisites named in the original article. Full original bibliography retained; all source-local definitions and supporting arguments retained.

1. Introduction

A pre-symplectic form is just a closed 2-form of constant rank. For instance, the restriction of a symplectic form to a coisotropic submanifold (such as the zero level set of a moment map) is pre-symplectic. Given a pre-symplectic from η of rank k, we constructed in [7] an algebraic structure that encodes the deformations of η, i.e., the 2-forms nearby η (in the C^0-sense) which are both closed and of constant rank k. As in many deformation problems, this algebraic structure is an L_∞-algebra, which we call the *Koszul L_∞-algebra* of η. Its construction – which is somewhat involved due to the simultaneous presence of the closedness and constant rank conditions – relies on a certain BV_∞-algebra structure on the differential forms and builds on the work of Fiorenza–Manetti [1]. The Koszul L_∞-algebra has the property that its Maurer–Cartan elements are in bijection with the pre-symplectic deformations of η.

Given that pre-symplectic forms are geometric objects, it is natural to ask for a geometric derivation of the algebraic structure that governs their deformations (the Koszul L_∞-algebra). The present note provides an answer to this question. The idea is the following: instead of restricting oneself to the realm of 2-forms, work in the larger class of almost Dirac structures, and consider deformations of

$$\operatorname{graph}(\eta) := \{(v, \eta(v,\cdot)) \,|\, v \in TM\} \subset TM \oplus T^*M$$

within the Dirac structures satisfying a constant rank condition. This is explained in Section 3.2, which is the heart of this note.

The first step in [7] is to provide a parametrization of the constant rank forms nearby η in terms of (an open subset in) a vector space. This parametrization is obtained naturally by taking the point of view of Dirac linear algebra in Section 3.3.

The second step in [7] was to show that the closedness condition translates into a Maurer–Cartan equation for a suitable L_∞-algebra. In Section 3.4 we re-obtain this result, and further we improve slightly a result of [7], see our Corollary 2.9.

Combining these results, in Section 3.5 we recover the fact that the L_∞-algebra governing deformations of Dirac structures, in the case at hand and upon a suitable restriction, is the Koszul L_∞-algebra.

The Koszul L_∞-algebra depends on an auxiliary choice of a distribution transverse to $\ker(\eta)$. In the Dirac-geometric interpretation, this translates into a suitable choice of a complement of $\operatorname{graph}(\eta)$ in $TM \oplus T^*M$. One of the achievements of [3] is to establish a general framework to control the effects of changing the complement, exhibiting explicit canonical L_∞-isomorphisms between the corresponding

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L_∞-algebras. A consequence of this note and of [3] is that the Koszul L_∞-algebra of (M, η) is well-defined up to L_∞-isomorphisms.

2. Review: deformations of pre-symplectic structures

We review the results on deformations of pre-symplectic structures obtained in the first three sections of [7].

2.1. **Pre-symplectic structures.** Fix a smooth manifold M.

Definition 2.1. A 2-form η on M is called *pre-symplectic* if

1) η is closed,
2) the vector bundle map $\eta^\sharp \colon TM \to T^*M, v \mapsto \iota_v\eta = \eta(v, \cdot)$ has constant rank.

A *pre-symplectic manifold* is a pair (M, η) consisting of a manifold M and a pre-symplectic structure η on M. We denote the space of all pre-symplectic structures of rank k on M by $\mathsf{Pre\text{-}Sym}^k(M)$.

A pre-symplectic manifold (M, η) gives rise to a distribution

$$K := \ker\big(\eta^\sharp\big).$$

This distribution is involutive since η is closed, hence K is tangent to a foliation of M. Denote by $r \colon \Omega(M) \to \Gamma(\wedge K^*)$ the restriction map. We define the horizontal differential forms as the elements of

$$\Omega_{\mathrm{hor}}(M) := \ker(r).$$

They form a subcomplex of the de Rham complex $\Omega(M)$, since the de Rham differential commutes with the pullback of differential forms. The subcomplex $\Omega_{\mathrm{hor}}(M)$ is the multiplicative ideal of $\Omega(M)$ generated by $\Gamma(K^\circ)$, where $K^\circ \subset T^*M$ denotes the annihilator of K.

2.2. **A parametrization of constant rank 2-forms.** In this subsection we fix a finite-dimensional, real vector space V. Recall that a bivector $Z \in \wedge^2 V$ is encoded by the induced linear map

$$Z^\sharp \colon\ V^* \to V, \qquad \xi \mapsto \iota_\xi Z = Z(\xi, \cdot).$$

Define

$$\mathcal{I}_Z := \big\{\beta \in \wedge^2 V^* \colon \mathrm{id}_V + Z^\sharp\beta^\sharp \text{ is invertible}\big\},$$

an open neighborhood of the origin in $\wedge^2 V^*$. Let $F \colon \mathcal{I}_Z \to \wedge^2 V^*$ be the map determined by

$$(F(\beta))^\sharp = \beta^\sharp\big(\mathrm{id} + Z^\sharp\beta^\sharp\big)^{-1}. \tag{2.1}$$

The map F is non-linear and smooth. It is a diffeomorphism from $\mathcal{I}_Z$ to $\mathcal{I}_{-Z}$, which keeps the origin fixed.

Fix $\eta \in \wedge^2 V^*$ of rank k. We now use F to construct submanifold charts for the space $\big(\wedge^2 V^*\big)_k$ of skew-symmetric bilinear forms on V of rank k. Fix a subspace $G \subset V$ such that $K \oplus G = V$, where $K = \ker\big(\eta^\sharp\big)$. The restriction of η to G is a non-degenerate skew bilinear form, therefore there is a unique $Z \in \wedge^2 G \subset \wedge^2 V$ such that

$$Z^\sharp \colon\ G^* \to G, \qquad \xi \mapsto \iota_\xi Z = Z(\xi, \cdot)$$

equals $-\big(\eta|_G^\sharp\big)^{-1}$.

Definition 2.2. The *Dirac exponential map* $\exp_\eta$ of η (and for fixed G) is the mapping

$$\exp_\eta\colon\ \mathcal{I}_Z \to \wedge^2 V^*, \qquad \beta \mapsto \eta + F(\beta).$$

Let $r\colon \wedge^2 V^* \to \wedge^2 K^*$ be the restriction map; we have the natural identification $\ker(r) \cong \wedge^2 G^* \oplus (G^* \otimes K^*)$. By the following theorem [7, Theorem 2.6], the restriction of $\exp_\eta$ to $\ker(r)$ is a submanifold chart for $\left(\wedge^2 V^*\right)_k \subset \wedge^2 V^*$.

Theorem 2.3 (parametrizing constant rank forms)**.**

(*i*) *Let $\beta \in \mathcal{I}_Z$. Then $\exp_\eta(\beta)$ lies in $\left(\wedge^2 V^*\right)_k$ if, and only if, β lies in $\ker(r) = (K^* \otimes G^*) \oplus \wedge^2 G^*$.*

(*ii*) *Let $\beta = (\mu, \sigma) \in \mathcal{I}_Z \cap \left((K^* \otimes G^*) \oplus \wedge^2 G^*\right)$. Then $\exp_\eta(\beta)$ is the unique skew-symmetric bilinear form on V with the following properties:*

- *its restriction to G equals $(\eta + F(\sigma))|_{\wedge^2 G}$;*
- *its kernel is the graph of the map $Z^\sharp \mu^\sharp = -\left(\eta|_G^\sharp\right)^{-1} \mu^\sharp\colon K \to G$.*

(*iii*) *The Dirac exponential map $\exp_\eta\colon \mathcal{I}_Z \to \wedge^2 V^*$ restricts to a diffeomorphism*

$$\mathcal{I}_Z \cap \left((K^* \otimes G^*) \oplus \wedge^2 G^*\right) \xrightarrow{\cong} \left\{\eta' \in \left(\wedge^2 V^*\right)_k \mid \ker\left((\eta')^\sharp\right) \text{ is transverse to } G\right\}$$

onto an open neighborhood of η in $\left(\wedge^2 V^*\right)_k$.

Remark 2.4. By the above linear algebra construction, given a pre-symplectic manifold (M, η), choosing a subbundle G complementary to $K = \ker\left(\eta^\sharp\right)$, one obtains a map

$$\exp_\eta\colon\ \mathcal{I}_Z \cap \left((K^* \otimes G^*) \oplus \left(\wedge^2 G^*\right)\right) \to \wedge^2 T^* M. \tag{2.2}$$

It is a not a vector bundle morphism but just a smooth fiberwise map. It maps the zero section to η, and its image is an open neighborhood of η in the space of 2-forms having the same rank as η. The map $\exp_\eta$ allows to parametrize deformations of η inside $\mathsf{Pre\text{-}Sym}^k(M)$ by means of sections $(\mu, \sigma) \in \Gamma(K^* \otimes G^*) \oplus \Gamma\left(\wedge^2 G^*\right) \cong \Omega^2_{\mathrm{hor}}(M)$ which are sufficiently small in the C^0-sense and for which the 2-form $(\exp_\eta)(\mu, \sigma)$ is a closed.

2.3. **An L_∞-algebra associated to a bivector field.** In this subsection we canonically associate an L_∞-algebra to any bivector field Z on a manifold M.

Definition 2.5. Let Z be a bivector field on M. The *Koszul bracket* associated to Z is the operation

$$[\cdot,\cdot]_Z\colon\ \Omega^r(M) \times \Omega^s(M) \to \Omega^{r+s-1}(M),$$
$$[\alpha, \beta]_Z := (-1)^{|\alpha|+1}\big(\mathcal{L}_Z(\alpha \wedge \beta) - \mathcal{L}_Z(\alpha) \wedge \beta - (-1)^{|\alpha|}\alpha \wedge \mathcal{L}_Z(\beta)\big).$$

Here $\mathcal{L}_Z = \iota_Z \circ d - d \circ \iota_Z$, where ι_Z denotes contraction with Z and d is the de Rham differential. On 1-forms α and β, the Koszul bracket reads $[\alpha, \beta]_Z = \mathcal{L}_{Z^\sharp \alpha}\beta - \mathcal{L}_{Z^\sharp \beta}\alpha - d\langle Z, \alpha \wedge \beta\rangle$.

In general the Koszul bracket of Z does not satisfy the graded Jacobi identity (it does only when Z is a Poisson bivector-field). We will see in Proposition 2.7 that nevertheless there is a well-behaved algebraic structure associated to Z. To this aim, recall that a differential form $\alpha \in \Omega^r(M)$ induces by contraction a linear map

$$\alpha^\sharp\colon\ TM \to \wedge^{r-1} T^* M, \qquad v \mapsto \iota_v \alpha,$$

and, following [2, Section 2.3], we extend this definition to a collection of forms $\alpha_1, \ldots, \alpha_n$ by setting

$$\alpha_1^\sharp \wedge \cdots \wedge \alpha_n^\sharp\colon\ \ \wedge^n TM \to \wedge^{|\alpha_1|+\cdots+|\alpha_n|-n} T^* M,$$
$$v_1 \wedge \cdots \wedge v_n \mapsto \sum_{\sigma \in S_n} (-1)^{|\sigma|} \alpha_1^\sharp(v_{\sigma(1)}) \wedge \cdots \wedge \alpha_n^\sharp(v_{\sigma(n)}).$$

Definition 2.6. We define the trinary bracket $[\cdot,\cdot,\cdot]_Z\colon \Omega^r(M)\times\Omega^s(M)\times\Omega^k(M)\to \Omega^{r+s+k-3}(M)$ associated to the bivector field Z to be

$$[\alpha,\beta,\gamma]_Z := \big(\alpha^\sharp\wedge\beta^\sharp\wedge\gamma^\sharp\big)\big(\tfrac{1}{2}[Z,Z]\big).$$

These brackets endow $\Omega(M)[2]$ with an $L_\infty[1]$-algebra structure, extending the results of Fiorenza and Manetti [5]. The following is [7, Proposition 3.5]:

Proposition 2.7 (the $L_\infty[1]$-algebra $\Omega(M)[2]$)**.** *Let Z be a bivector field on M. The multilinear maps λ_1, λ_2, λ_3 on the graded vector space $\Omega(M)[2]$ given by*

1) *λ_1 the de Rham differential d,*
2) *$\lambda_2(\alpha[2]\odot\beta[2]) = -\big(\mathcal{L}_Z(\alpha\wedge\beta) - \mathcal{L}_Z(\alpha)\wedge\beta - (-1)^{|\alpha|}\alpha\wedge\mathcal{L}_Z(\beta)\big)[2] = (-1)^{|\alpha|}([\alpha,\beta]_Z)[2]$, and*
3) *$\lambda_3(\alpha[2]\odot\beta[2]\odot\gamma[2]) = (-1)^{|\beta|+1}\big(\alpha^\sharp\wedge\beta^\sharp\wedge\gamma^\sharp\big(\tfrac{1}{2}[Z,Z]\big)\big)[2]$,*

define the structure of an $L_\infty[1]$-algebra on $\Omega(M)[2]$.

We now explain the geometric relevance of the $L_\infty[1]$-algebra $(\Omega(M)[2],\lambda_1,\lambda_2,\lambda_3)$. As for any $L_\infty[1]$-algebra, it comes with distinguished elements:

Definition 2.8. An element $\beta\in\Omega^2(M)$ is a *Maurer–Cartan element* of $(\Omega(M)[2],\lambda_1,\lambda_2,\lambda_3)$ if it satisfies the *Maurer–Cartan equation*

$$d(\beta[2]) + \tfrac{1}{2}\lambda_2(\beta[2]\odot\beta[2]) + \tfrac{1}{6}\lambda_3(\beta[2]\odot\beta[2]\odot\beta[2]) = 0.$$

Recall that at the beginning of Section 2.2 we defined an open subset $\mathcal{I}_Z\subset\wedge^2T^*M$ and a map $F\colon\mathcal{I}_Z\to\wedge^2T^*M$. The following is [7, Corollary 3.9].

Corollary 2.9 (Maurer–Cartan elements of $\Omega(M)[2]$)**.** *There is an open subset $\mathcal{U}\subset\mathcal{I}_Z$, which contains the zero section of $\wedge^2T^*M$, such that a 2-form $\beta\in\Gamma(\mathcal{U})$ is a Maurer–Cartan element of $(\Omega(M)[2],\lambda_1,\lambda_2,\lambda_3)$ if, and only if, the 2-form $F(\beta)$ is closed.*

In Section 3.4 we will show that for the open subset $\mathcal{U}$ one can choose the whole of $\mathcal{I}_Z$.

2.4. **The Koszul L_∞-algebra of a pre-symplectic manifold.** Let again η be a pre-symplectic structure on a manifold M. Fix a subbundle $G\subset TM$ which is complementary to the kernel K of η. Consider the bivector field Z satisfying $Z^\sharp = -\big(\eta|_G^\sharp\big)^{-1}$. The following is [7, Theorem 3.17].

Theorem 2.10 (the Koszul $L_\infty[1]$-algebra)**.** *The $L_\infty[1]$-algebra structure on $\Omega(M)[2]$ associated to the bivector field Z, see Proposition 2.7, maps $\Omega_{\mathrm{hor}}(M)[2]$ to itself. The subcomplex $\Omega_{\mathrm{hor}}(M)[2]\subset\Omega(M)[2]$ therefore inherits the structure of an $L_\infty[1]$-algebra, which we call the Koszul $L_\infty[1]$-algebra of (M,η).*

We denote by $\mathsf{MC}(\eta)$ the set of Maurer–Cartan elements of the Koszul $L_\infty[1]$-algebra of (M,η).

In view of the above theorem, the following result [7, Theorem 3.19] is an immediate consequence of Theorem 2.3 and Corollary 2.9.

Theorem 2.11 (Maurer–Cartan elements of the Koszul $L_\infty[1]$-algebra)**.** *Let (M,η) be a pre-symplectic manifold. The choice of a complement G to the kernel of η determines a bivector field Z by requiring $Z^\sharp = -\big(\eta|_G^\sharp\big)^{-1}$. Suppose β is a 2-form on M, which lies in $\mathcal{I}_Z$. The following statements are equivalent:*

1. *β is a Maurer–Cartan element of the Koszul $L_\infty[1]$-algebra $\Omega_{\mathrm{hor}}(M)[2]$ of (M,η), which was introduced in Theorem 2.10.*
2. *The image of β under the map $\exp_\eta$, which is introduced in Definition 2.2, is a pre-symplectic structure of the same rank as η.*

The above Theorem 2.11 is the main result of [7], as it states that the Koszul $L_\infty[1]$-algebra governs the deformations of the pre-symplectic structure η. More precisely: the fibrewise map $\exp_\eta$ as in (2.2), on the level of sections, restricts to a map

$$\boxed{\exp_\eta\colon\ \Gamma(\mathcal{I}_Z)\cap \mathsf{MC}(\eta)\to \mathsf{Pre\text{-}Sym}^k(M)}$$

which is injective and whose image consists of the pre-symplectic structures of rank equal to the rank of η and with kernel transverse to G.

3. Dirac geometric interpretation

In the remainder of this note we explain the geometric framework that underlies the results of Section 2 recalled from [7]. We recover naturally the statements made there and provide some alternative and more geometric proofs.

3.1. **Background on Dirac geometry.** We first review some notions from Dirac linear algebra. Let V be a finite-dimensional, real vector space. We denote by $\mathbb{V}$ the direct sum $V\oplus V^*$ and by $\langle\cdot,\cdot\rangle$ the following non-degenerate pairing on $\mathbb{V}$:

$$\langle(v,\xi),(w,\chi)\rangle := \xi(w)+\chi(v).$$

Definition 3.1. A subspace $W\subset\mathbb{V}$ is called *Lagrangian* if for all $w,w'\in W$ we have $\langle w,w'\rangle=0$ and $\dim(W)=\dim(V)$. Two subspaces W and $\tilde W\subset\mathbb{V}$ are *transverse*, if $W\oplus\tilde W=\mathbb{V}$.

Given an element $Z\in\wedge^2V$, we defined the linear map $Z^\sharp\colon V^*\to V$ in Section 2.2, and we can consider the Lagrangian subspace $\mathrm{graph}(Z):=\left\{\left(Z^\sharp\xi,\xi\right)\,|\,\xi\in V^*\right\}\subset\mathbb{V}$. Similarly, for $\beta\in\wedge^2V^*$ we define $\beta^\sharp\colon V\to V^*$ and consider $\mathrm{graph}(\beta)$.

Every $\beta\in\wedge^2V^*$ defines an orthogonal transformation $\mathfrak{t}_\beta$ of $(\mathbb{V},\langle\cdot,\cdot\rangle)$, by

$$(v,\xi)\mapsto\big(v,\xi+\beta^\sharp(v)\big).$$

Similarly, every $Z\in\wedge^2V$ gives rise to an orthogonal transformation $\mathfrak{t}_Z$, which takes (v,ξ) to $\big(v+Z^\sharp(\xi),\xi\big)$. In particular, elements of $\wedge^2V^*$ and $\wedge^2V$ act on the set of Lagrangian subspaces of $\mathbb{V}$.

Remark 3.2. Suppose L, R are transverse Lagrangian subspaces of $\mathbb{V}$. There is a canonical isomorphism

$$R\cong L^*,\qquad r\mapsto\langle r,\cdot\rangle|_L.$$

Since R is transverse to L, any subspace of $\mathbb{V}$ transverse to R is the graph of a linear map $L\to R$. Any *Lagrangian* subspace transverse to R is the graph of a linear map $L\to R$ such that, composing with the canonical isomorphism above, we obtain a *skew-symmetric* linear map $L\to L^*$ (i.e., the sharp map associated to an element of $\wedge^2L^*$).

Let us now briefly recall the basic constituencies of Dirac geometry. Consider the generalized tangent bundle $\mathbb{T}M=TM\oplus T^*M$. It comes equipped with a non-degenerate pairing

$$\langle(X,\alpha),(Y,\beta)\rangle:=\alpha(Y)+\beta(X)$$

and the Dorfman bracket

$$[\![(X,\alpha),(Y,\beta)]\!]=([X,Y],\mathcal{L}_X\beta-\iota_Yd\alpha).$$

Together with the projection to TM, this makes $\mathbb{T}M$ into an example of Courant algebroid.

Definition 3.3. An *almost Dirac structure* on M is a Lagrangian subbundle $L \subset (\mathbb{T}M, \langle\cdot,\cdot\rangle)$. A *Dirac structure* is an almost Dirac structure whose space of sections is closed with respect to the Dorfman bracket $[\![\cdot,\cdot]\!]$.

Remark 3.4. Let L, R be transverse Dirac structures on M. As seen in Remark 3.2, almost Dirac structures transverse to R are in bijection with elements of $\Gamma\big(\wedge^2 L^*\big)$. We now recall a result of Liu–Weinstein–Xu [4] establishing when such an almost Dirac structure is Dirac. Recall that every Dirac structure, with the restricted Dorfman bracket and anchor, is a Lie algebroid. Since L is a Lie algebroid, it induces a differential d_L on $\Gamma(\wedge L^*)$. Further[1], since $L^* \cong R$ is a Lie algebroid, it induces a graded Lie bracket $[\cdot,\cdot]_{L^*}$ on $\Gamma(\wedge L^*)[1]$. Together with d_L and $[\cdot,\cdot]_{L^*}$, the graded vector space $\Gamma(\wedge L^*)[1]$ becomes a differential graded Lie algebra. The main result of [4] is: for all $\varepsilon \in \Gamma\big(\wedge^2 L^*\big)$, the graph $L_\varepsilon = \{v + \iota_v\varepsilon \colon v \in L\}$ is a Dirac structure iff ε satisfies the Maurer–Cartan equation, that is

$$d_L\varepsilon + \tfrac{1}{2}[\varepsilon,\varepsilon]_{L^*} = 0.$$

3.2. **Deformations of pre-symplectic structures: the point of view of Dirac geometry.** In this subsection we cast the deformations of pre-symplectic forms in the framework of Dirac geometry.

Let η be a pre-symplectic form on M, with kernel K. The natural way to parametrize deformations of η is by 2-forms α such that $\eta+\alpha$ is again pre-symplectic, but this parametrization has a serious flaw: the space of such α's does not have a natural vector space structure, due to the constant rank condition. Taking the point of view of Dirac geometry, the above approach to parametrize the deformations of η amounts to deforming the Dirac structure $\mathrm{graph}(\eta)$ using $\{0\} \oplus T^*M$ as a complement.

A better way to parametrize the deformations of η in terms of Dirac geometry works as follows. Let us first choose a complement G to K. Then

$$G \oplus K^*$$

is a complement[2] of $\mathrm{graph}(\eta)$. We can now use $G \oplus K^*$ – instead of $\{0\} \oplus T^*M$ – to parametrize deformations of the Dirac structure $\mathrm{graph}(\eta)$. This choice of complement has the advantage of linearizing the constant rank condition, as we show in Proposition 3.7 below. (Notice that when η is *symplectic*, the new complement is just TM, hence we are deforming η by viewing it as a Poisson structure, just as in [7, Section 1.3].)

We first state two lemmas about the effect of applying the orthogonal transformation $\mathfrak{t}_{-\eta}$ of $TM \oplus T^*M$, given by $(v,\xi) \mapsto \big(v, \xi - \eta^\sharp(v)\big)$.

Lemma 3.5. *Denote by $Z \in \Gamma\big(\wedge^2 G\big)$ the bivector field such that $Z^\sharp$ is the inverse of $-(\eta|_G)^\sharp$. Then $\mathfrak{t}_{-\eta}$ maps $G \oplus K^*$ to $\mathrm{graph}(Z)$.*

Proof. $\mathfrak{t}_\eta(\mathrm{graph}(Z)) = \big\{\big(Z^\sharp\xi, \xi|_K\big) \colon \xi \in T^*M\big\} = G \oplus K^*$. □

Lagrangian subbundles nearby $\mathrm{graph}(\eta)$ can be written, for some $\bar\beta \in \Gamma(\wedge^2(\mathrm{graph}(\eta))^*)$, as the graph of the map

$$\bar\beta^\sharp \colon \ \mathrm{graph}(\eta) \to (\mathrm{graph}(\eta))^* \cong G \oplus K^*,$$

by Remark 3.2. We denote this graph as $\Phi_{G\oplus K^*}\big(\bar\beta\big)$. Moreover, let $\beta \in \Omega^2(M)$ be the 2-form corresponding to $\bar\beta$ under the isomorphism $\mathrm{graph}(\eta) \cong TM, v + \iota_v\eta \mapsto v$ and denote by $\Phi_Z(\beta)$ the graph of the map $\beta^\sharp \colon TM \to T^*M \cong \mathrm{graph}(Z)$.

[1]The Lie algebroid structures on L and L^* are compatible in the sense that the pair (L, L^*) forms a Lie bialgebroid.

[2]Indeed, for every $v \in TM$ we have $\iota_v\eta \in K^\circ = G^*$, so requiring that $\iota_v\eta$ lies in K^* implies $\iota_v\eta = 0$. This means that $v \in K$, so requiring that v lies in G implies $v = 0$.

Lemma 3.6. $\mathfrak{t}_{-\eta}$ *maps* $\Phi_{G\oplus K^*}\big(\bar\beta\big)$ *to* $\Phi_Z(\beta)$.

Proof. $\mathfrak{t}_{-\eta}$ preserves the pairing on $TM\oplus T^*M$, clearly maps $\mathrm{graph}(\eta)$ to TM, and maps $G\oplus K^*$ to $\mathrm{graph}(Z)$ by Lemma 3.5. Therefore the statement follows by functoriality. □

Now we can explain why the choice of $G\oplus K^*$ as a complement is a good one to describe pre-symplectic deformations.

Proposition 3.7. *Let* $\bar\beta\in\Gamma\big(\wedge^2(\mathrm{graph}(\eta))^*\big)$.

(*i*) *The rank of*

$$\Phi_{G\oplus K^*}\big(\bar\beta\big)\cap TM \tag{3.1}$$

equals the rank of

$$\{v\in K\colon \iota_v\beta\in G^*\}. \tag{3.2}$$

(*ii*) *Assume that* $\Phi_{G\oplus K^*}\big(\bar\beta\big)$ *is the graph of a 2-form. Then the rank of this 2-form equals* $\mathrm{rank}(\eta)$ *iff* β *lies in the vector space* $\Omega^2_{\mathrm{hor}}(M)$ *of horizontal 2-forms.*

Proof. (i) Applying the transformation $\mathfrak{t}_{-Z}\circ\mathfrak{t}_{-\eta}$ to $\Phi_{G\oplus K^*}\big(\bar\beta\big)$, by Lemma 3.6 we obtain $\mathfrak{t}_{-Z}(\Phi_Z(\beta))=\mathrm{graph}(\beta)$. Applying it to TM we obtain $\big\{\big(v+Z^\sharp\iota_v\eta,-\iota_v\eta\big)\,|\,v\in TM\big\}=K\oplus G^*$.

Hence applying the transformation to the intersection (3.1) we obtain

$$\mathrm{graph}(\beta)\cap(K\oplus G^*),$$

which is isomorphic to (3.2).

(ii) Denote by η' the 2-form whose graph is $\Phi_{G\oplus K^*}\big(\bar\beta\big)$. The kernel of η' is given by (3.1), and the assertion follows immediately from (i). Recall that the vector space $\Omega^2_{\mathrm{hor}}(M)$ of horizontal 2-forms was defined in Section 2.1, as the space of 2-forms that vanish on $\wedge^2K$. □

Remark 3.8. Since $\mathfrak{t}_{-\eta}$ is actually an automorphism of the standard Courant algebroid $TM\oplus T^*M$, the following two deformation problems of Dirac structures are equivalent:

- deformations of $\mathrm{graph}(\eta)$, using the complement $G\oplus K^*$,
- deformations of TM, using the complement $\mathrm{graph}(Z)$.

The latter deformation problem is easier to handle, and the $L_\infty[1]$-algebra structure governing it will be recovered in Section 3.4.

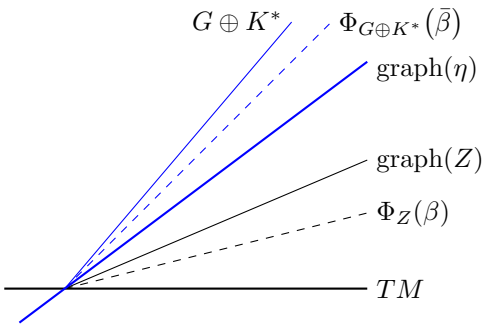


FIGURE 1. The Dirac structures $\mathrm{graph}(\eta)$ and TM, together with the complementary Lagrangian subbundles we use to deform them.

3.3. **Dirac-geometric interpretation of Section 2.2.** Using Dirac linear algebra, we explain and re-prove the results recalled in Section 2.2.

3.3.1. *Revisiting the map* $\boldsymbol{F}$ *from formula (2.1).* Let V be a finite-dimensional, real vector space. We fix a bivector $Z \in \wedge^2 V$. Recall that $\mathcal{I}_Z$ consists of elements $\beta \in \wedge^2 V^*$ such that $\mathrm{id} + Z^\sharp \beta^\sharp$ is invertible. In formula (2.1), we defined the map $F\colon \mathcal{I}_Z \to \wedge^2 V^*$ given by

$$(F(\beta))^\sharp = \beta^\sharp \big(\mathrm{id} + Z^\sharp \beta^\sharp\big)^{-1}.$$

The following lemma provides a geometric explanation of the map F.

Lemma 3.9. *Fix $Z \in \wedge^2 V$.*

(*i*) *Taking graphs with respect to the decompositions $\mathbb{V} = V \oplus V^*$ resp. $\mathbb{V} = V \oplus \mathrm{graph}(Z)$, yields bijections*

$$\begin{aligned} \Phi_0\colon \wedge^2 V^* &\xrightarrow{\cong} \{\textit{Lagrangian subspace of } \mathbb{V} \textit{ transverse to } V^*\},\\ \alpha &\mapsto \{(v, \iota_v \alpha) \,|\, v \in V\},\\ \Phi_Z\colon \wedge^2 V^* &\xrightarrow{\cong} \{\textit{Lagrangian subspace of } \mathbb{V} \textit{ transverse to } \mathrm{graph}(Z)\},\\ \beta &\mapsto \big\{\big(v + Z^\sharp(\iota_v\beta), \iota_v\beta\big) \,|\, v \in V\big\}. \end{aligned}$$

(*ii*) *Given $\beta \in \wedge^2 V^*$, the Lagrangian subspace $\Phi_Z(\beta)$ is transverse to $V^* \subset \mathbb{V}$ if, and only if $\beta \in \mathcal{I}_Z$.*

(*iii*) *The map*

$$\Phi_0^{-1} \circ \Phi_Z\colon \mathcal{I}_Z \to \wedge^2 V^*$$

is well-defined and coincides with F.

In particular, the map F is characterized by the property that

$$\mathrm{graph}(F(\beta)) = \Phi_Z(\beta) \tag{3.3}$$

for all $\beta \in \mathcal{I}_Z$. In other words, $F(\beta)$ is obtained taking the graph of β w.r.t. the splitting $\mathbb{V} = V \oplus \mathrm{graph}(Z)$.

Proof. (i) According to Remark 3.2, any Lagrangian subspace transverse to V^* is the graph of a skew-symmetric linear map $V \to V^*$, and therefore can be written as $\{(v, \iota_v\alpha) \,|\, v \in V\}$ for some $\alpha \in \wedge^2 V^*$. Similarly, $\mathrm{graph}(Z)$ is transverse to V and the induced isomorphism $\mathrm{graph}(Z) \cong V^*$ is just $(Z^\sharp(\xi), \xi) \mapsto \xi$. Hence any Lagrangian subspace transverse to $\mathrm{graph}(Z)$ can be written as $\{(v, 0) + (Z^\sharp(\iota_v\beta), \iota_v\beta) \,|\, v \in V\}$ for some $\beta \in \wedge^2 V^*$.

(ii) The expression for $\Phi_Z(\beta)$ in item (i) shows that $\Phi_Z(\beta) \cap V^* = \{(0, \iota_v\beta) \,|\, v \in V, v + Z^\sharp(\iota_v\beta) = 0\}$. This intersection is trivial iff $\ker\big(\mathrm{id} + Z^\sharp\beta^\sharp\big) \subseteq \ker\big(\beta^\sharp\big)$. In turn, this condition is equivalent to $\big(\mathrm{id} + Z^\sharp\beta^\sharp\big)$ being injective, and thus invertible.

(iii) Finally, if $\mathrm{id} + Z^\sharp\beta^\sharp$ is invertible, $\Phi_Z(\beta)$ is transverse to V^* by item (ii). By item (i) the element $\Phi_0^{-1}(\Phi_Z(\beta))$ is well-defined. In concrete terms, it is given by $\alpha \in \wedge^2 V^*$ such that for all $v \in V$, there is $w \in V$ for which

$$\big(v + Z^\sharp\beta^\sharp(v), \beta^\sharp(v)\big) = \big(w, \alpha^\sharp(w)\big)$$

holds. Equivalently, this means that $\alpha^\sharp\big(\mathrm{id} + Z^\sharp\beta^\sharp\big)(v) = \beta^\sharp(v)$ for all $v \in V$. This shows that $\Phi_0^{-1} \circ \Phi_Z$ agrees with F. □

3.3.2. *Revisiting Theorem 2.3 (parametrizing constant rank forms).* Now let $\eta \in \wedge^2 V^*$ be of rank k, fix a complement G to $K := \ker(\eta)$, and denote by $Z \in \wedge^2 G$ the bivector determined by $Z^\sharp = -\big(\eta|_G^\sharp\big)^{-1}$. In Section 3.2 we considered deformations of the Dirac structure $\mathrm{graph}(\eta)$ using $G \oplus K^*$ as a complement. They are graphs of 2-forms given by the Dirac exponential map $\exp_\eta$ (see Definition 2.2). More precisely:

Lemma 3.10. *For all $\beta \in \mathcal{I}_Z$ we have*

$$\mathrm{graph}(\exp_\eta(\beta)) = \Phi_{G\oplus K^*}\big(\bar\beta\big). \tag{3.4}$$

Proof. We have $\mathrm{graph}(\exp_\eta(\beta)) = \mathfrak{t}_\eta(\Phi_Z(\beta)) = \Phi_{G\oplus K^*}\big(\bar\beta\big)$, where the first equality holds by equation (3.3) and the second by Lemma 3.6. □

Using this we recover Theorem 2.3, in particular item (i) stating that $\exp_\eta(\beta)$ has rank equal to $k = \dim(K)$ iff β is horizontal.

Alternative proof of Theorem 2.3. (i) Apply Proposition 3.7(ii) together with equation (3.4).

(ii) We only prove the statement about the kernel of $\exp_\eta(\beta)$. Write $\beta = (\mu, \sigma)$. By the proof of Proposition 3.7(i), the intersection of the subspace (3.4) with V is $(\mathfrak{t}_\eta \circ \mathfrak{t}_Z)(\mathrm{graph}(\beta) \cap (K \oplus G^*))$, which is precisely the image of K under $\mathrm{id} + Z^\sharp\mu^\sharp$.

(iii) By Lemma 3.9(ii), the map Φ_Z provides a bijection between $\mathcal{I}_Z$ and Lagrangian subspaces transverse to $\mathrm{graph}(Z)$ and to V^*. Hence $\mathfrak{t}_\eta \circ \Phi_Z$ provides a bijection between $\mathcal{I}_Z$ and Lagrangian subspaces transverse to $\mathfrak{t}_\eta(\mathrm{graph}(Z)) = G \oplus K^*$ (see Lemma 3.5) and to V^*. The latter are exactly the graphs of elements $\eta' \in \wedge^2 V^*$ so that the $\eta'|_{\wedge^2 G}$ is non-degenerate. Hence, by the proof of Lemma 3.10, $\exp_\eta$ provides a bijection between $\mathcal{I}_Z$ and such η'. We conclude using (i). □

3.4. **Dirac-geometric interpretation of Section 2.3.** Using Dirac geometry and adapting results from [2], we explain and re-prove the results recalled in Section 2.3. Fix a bivector field Z on M.

3.4.1. *Revisiting Proposition 2.7 (the $\boldsymbol{L_\infty[1]}$-algebra $\boldsymbol{\Omega(M)[2]}$).* In Proposition 2.7, the $L_\infty[1]$-algebra $(\Omega(M)[2], \lambda_1, \lambda_2, \lambda_3)$ was constructed out of a bivector field Z. It can be recovered using Dirac geometry – or more precisely, the deformation theory of Dirac structures – as a special case of the construction from [2, Section 2.2].

Proposition 3.11. *Let L be a Dirac structure and R a complementary almost Dirac structure, i.e., we have a vector bundle decomposition $L \oplus R = \mathbb{T}M$. Then $\Gamma(\wedge L^*)[2]$ has an induced $L_\infty[1]$-algebra structure, whose only non-trivial multibrackets are μ_1, μ_2, μ_3 given as follows:*

1) *μ_1 is the differential d_L associated to the Lie algebroid L,*
2) *$\mu_2(\alpha[2] \odot \beta[2]) = -(-1)^{|\alpha|}[\alpha, \beta]_{L^*}[2]$, where $[\cdot,\cdot]_{L^*} := \mathrm{pr}_R([\![\cdot,\cdot]\!])$ denotes the (extension of) the bracket of the almost Lie algebroid $R \cong L^*$,*
3) *$\mu_3(\alpha[2] \odot \beta[2] \odot \gamma[2]) = (-1)^{|\beta|}\big(\alpha^\sharp \wedge \beta^\sharp \wedge \gamma^\sharp\big)\psi[2]$, where $\psi \in \Gamma\big(\wedge^3 L\big)$ is given by $\Gamma\big(\wedge^3 L^*\big) \to \mathcal{C}^\infty(M)$, $\xi_1 \wedge \xi_2 \wedge \xi_3 \mapsto \langle \mathrm{pr}_L([\![\xi_1, \xi_2]\!]), \xi_3\rangle$, where we made use of the identification $R \cong L^*$.*

More generally, Proposition 3.11 holds if replacing $\mathbb{T}M$ by any Courant algebroid.

Proof. The proof is a minor adaptation of the first part of the proof of [2, Lemma 2.6], setting $\varphi = 0$ there. We recall briefly the idea of the latter. By [6] there is a natural description of the Courant algebroid structure on $\mathbb{T}M$ in terms of graded geometry. One can use it to apply Voronov's higher derived brackets construction (see [8, 9]) and obtain an $L_\infty[1]$-algebra structure on $\Gamma(\wedge L^*)[2]$. The multibrackets

obtained are the ones in the statement of the lemma, as one checks using [6] and via computations in local coordinates. □

Alternative proof of Proposition 2.7. Let Z be a bivector field on M. We apply Proposition 3.11 for the case $L = TM$ and $R = \mathrm{graph}(Z)$. In this case d_L is the de Rham differential, and the bracket on R is given by the formula for the Koszul bracket. One checks that ψ is the trivector field $-\frac{1}{2}[Z, Z]$, using [7, Lemma 1.6]. Hence the $L_\infty[1]$-brackets on $\Omega(M)[2]$ given by Proposition 3.11 are $\mu_1 = \lambda_1$, $\mu_2 = -\lambda_2$ and $\mu_3 = \lambda_3$. Applying the automorphism $-\mathrm{id}$ to $\Omega(M)[2]$ yields Proposition 2.7. □

3.4.2. *Revisiting Corollary 2.9 (Maurer–Cartan elements of* $\boldsymbol{\Omega(M)[2]}$*).* We now turn to Maurer–Cartan elements. In Lemma 3.9(i), we gave a parametrization of all almost Dirac structures that are transverse to $\mathrm{graph}(Z)$ in terms of 2-forms β on M. This parametrization is given by

$$\beta \mapsto \Phi_Z(\beta) = \big\{ \big(v + Z^\sharp(\iota_v\beta), \iota_v\beta\big) \,|\, v \in TM \big\}.$$

We present the second part of [2, Lemma 2.6], which is an extension of the work by Liu–Weinstein–Xu recalled in Remark 3.4.

Proposition 3.12. *Let L be given a Dirac structure and R a complementary almost Dirac structure. An element $\sigma \in \Gamma\big(\wedge^2 L^*\big)[2]$ is a Maurer–Cartan element of the $L_\infty[1]$-algebra structure given in Proposition* 3.11 *iff the graph*

$$\Gamma_\sigma := \{(X - \iota_X\sigma) \colon X \in L\} \subset L \oplus R$$

is a Dirac structure. (The above inclusion makes use of the identification $R \cong L^*$.)

Corollary 2.9 states that for $\beta \in \Omega^2(M)$ taking values in some sufficiently small neighborhood $\mathcal{U}$ of the zero section in $\wedge^2 T^*M$ – in particular taking values in $\mathcal{I}_Z$, i.e., $\mathrm{id} + Z^\sharp\beta^\sharp$ is invertible, – β is a Maurer–Cartan element of $(\Omega(M)[2], \lambda_1, \lambda_2, \lambda_3)$ iff $F(\beta)$ is closed. We now provide an alternative proof of this result, which also shows that one can choose $\mathcal{U}$ to equal $\mathcal{I}_Z$.

Alternative proof of Corollary 2.9. For any $\beta \in \Omega^2(M)$, being a Maurer–Cartan element of the $L_\infty[1]$-algebra $(\Omega(M)[2], \lambda_1, \lambda_2, \lambda_3)$ is equivalent to $\Phi_Z(\beta)$ being a Dirac structure. This follows from applying Proposition 3.12 to the Dirac structure $L = TM$ and to the almost Dirac structure $R = \mathrm{graph}(Z)$, noticing that $\Gamma_{-\beta} = \{(v + Z^\sharp(\iota_v\beta), \iota_v\beta) \,|\, v \in TM\} = \Phi_Z(\beta)$. When $\beta \in \Gamma(\mathcal{I}_Z)$, we know that $\Phi_Z(\beta)$ can be written as the graph of the 2-form $F(\beta)$, by equation (3.3). Now use the fact that the graph of a 2-form is a Dirac structure if, and only if, the 2-form is closed. □

Remark 3.13. In this subsection we recovered the $L_\infty[1]$-algebra $\Omega(M)[2]$ of Proposition 2.7 as the $L_\infty[1]$-algebra governing deformations of the Dirac structure TM taking $\mathrm{graph}(Z)$ as a complement. By Remark 3.8, this deformation problem is equivalent to the deformations of the Dirac structure $\mathrm{graph}(\eta)$ taking $G \oplus K^*$ as the complement. This explains why the $L_\infty[1]$-algebra $\Omega(M)[2]$ governs the latter deformation problem, and therefore is relevant for the deformations of pre-symplectic structures.

3.5. **Dirac-geometric interpretation of Section 2.4.** Theorem 2.10 can be deduced from a general statement about (almost) Dirac structures, however doing so amounts essentially to the same computations that were needed for the proof given in [7]. We include this general statement for the sake of completeness.

Proposition 3.14. *In the setting of Proposition* 3.11*, let K be a subbundle of L and define $\Gamma_{\mathrm{hor}}(\wedge L^*)$ as the kernel of the restriction map $\Gamma(\wedge L^*) \to \Gamma(\wedge K^*)$. Then the multibrackets μ_1, μ_2, μ_3 preserve $\Gamma_{\mathrm{hor}}(\wedge L^*)[2]$ iff K satisfies the following:*

- *K is a Lie subalgebroid of L,*
- *$\langle [\![\xi_1, \xi_2]\!], K + K^\circ \rangle = 0$ for all $\xi_1, \xi_2 \in \Gamma(K^\circ)$, where we use the identification $K^\circ \subset L^* \cong R$ and $[\![\cdot, \cdot]\!]$ denotes the Dorfman bracket.*

Proof. We will use the fact that μ_1, μ_2, μ_3 are derivations w.r.t. the wedge product in each entry. The Lie algebroid differential d_L preserves $\Gamma_{\mathrm{hor}}(\wedge L^*)$ iff the subbundle K is involutive. The bracket $[\cdot, \cdot]_{L^*}$ preserves $\Gamma_{\mathrm{hor}}(\wedge L^*)$ iff $\langle [\![\xi_1, \xi_2]\!], K \rangle = 0$ for all $\xi_1, \xi_2 \in \Gamma(K^\circ)$. The trinary bracket μ_3 preserves $\Gamma_{\mathrm{hor}}(\wedge L^*)$ iff $\mu_3(\xi_1, \xi_2, \xi_3) = 0$ for all $\xi_i \in \Gamma(K^\circ)$, which in turn is equivalent to $\langle [\![\xi_1, \xi_2]\!], \xi_3 \rangle = 0$. □

Finally, as mentioned earlier, Theorem 2.11 follows immediately from the other results presented.

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